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Assignment

Q1 What do you understand by image? What are different types of images? What are the different colour channels in which the images can be represented?

Ans An image is a visual representation of an object, scene or information in digital form. A digital image is made up of tiny picture elements called pixels, where each pixel stores intensity or colour information. Together, these pixels form the complete image.

→ Different types of digital images are :-

1. **Binary Image** :-

- Contains only two pixel values: 0 (black) and 1 (white).
- Used in scanned documents, QR codes, and text recognition.

2. **Greyscale Image** :-

- Contains shades of gray ranging from black (0) to white (255).
- Each pixel stores only intensity information.

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5. Colour Image:-

- Represents images using multiple colour channels, most commonly RGB.
- Used in digital cameras, smartphones, and displays.

4. RGBA Image:-

- Similar to RGB but includes an Alpha channel for transparency.
- Common in graphics, logos, and web design.

→ Different colour channels in which images can be represented are:-

1. RGB (Red, Green, Blue):- These channels used in computer monitors, TVs, and cameras.
2. RGBA (Red, Green, Blue, Alpha): RGB plus an Alpha channel for transparency.
3. CMYK (Cyan, Magenta, Yellow, Black): Used in colour printing.
4. HSV (Hue, Saturation, Value): Represents colors based on hue, saturation, and brightness; useful in image processing.

HSL

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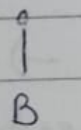
5. YCbCr (Luminance, Blue Chrominance, Red Chrominance) :-

- Commonly used in image and video compression (e.g., JPEG, MPEG).

Image Repⁿ

1) Binary (0,1)
2) Grayscale (0,255)
3) Colour $2^8 \rightarrow 256$

RGB - (0-255)



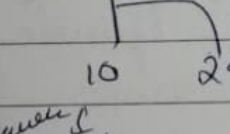
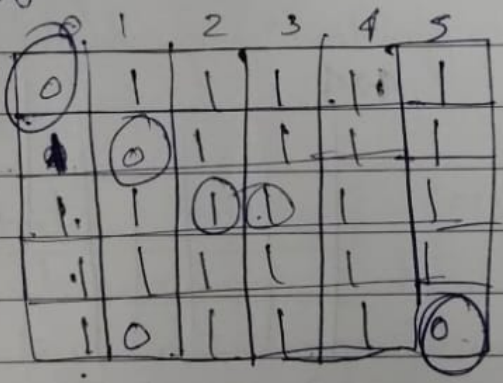


Image Processing



DIP Digital Image Processing

When an image is converted in the form of 0s & 1s or has been digitalized.

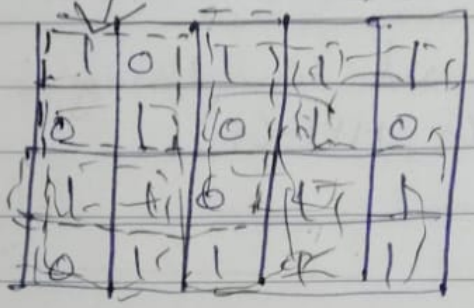
Coordinates of an image → (x, y)

(0,0)
(4,5)

Intensity of ~~an pixels~~ image

f(x,y) → the value at a particular loc^n representing Grayscale value

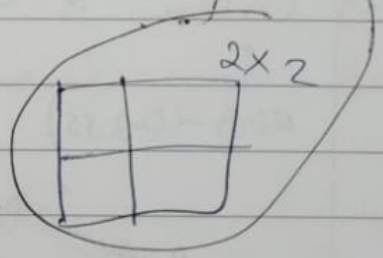
Q What do you understand by sampling & quantization of an image?

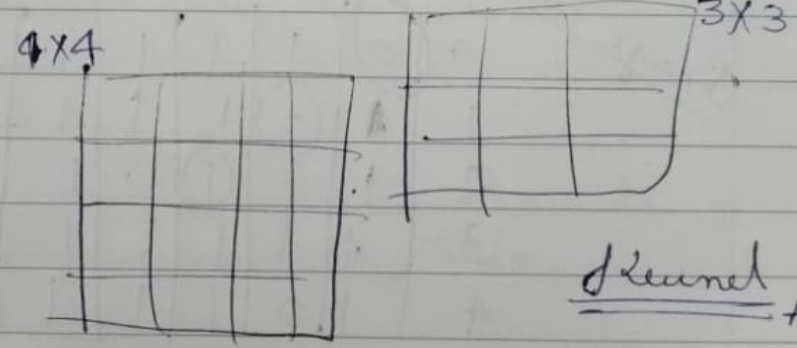


Filtering → Filtering means removing noise, enhancing image, etc. to extract some meaningful info. by applying some filters.

Filtering
or
Filter
or
Mask
or
Kernel

→ noise





Kernel / Filter

→ Small Img → Small Kernel
→ Large —11— Large —11—

Two types :-

1) Special domain filtering
2) Freq. domain filtering

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Names of filters:-

1. **Smoothening filters** :- These filters are used to reduce noise & soften image texture. There are two types
   i) Mean Filter
   ii) Gaussian Filter

2. **Sharpening Filters**:- These filters are used to highlight fine details & enhance edges by emphasizing abrupt transactions in pixel intensity. Two types:-
   i) Laplacian Filter
   ii) Unsharp Masking

   (for technical depth)

3. **Edge Detection Filters** :- These filters highlights boundaries where obj.s & backgrounds separate by detecting sharp gradients in the image.
   i) Sobel Filter
   ii) Canny edge detector

4. **Noise Reduction** :- These filters target specific kind of noise w/o excessively degrading the overall image.
   i) Median Filter

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5. **Frequency domain filters**:- These operate by transforming an image into its freq. components using the fourier transform.
   i) Low Pass Filter
   ii) Highpass Filter

6. **Morphological filters**:- These filters utilizes structural elements to alter the geometric structure of an image.
   i) Erosion
   ii) Dilation

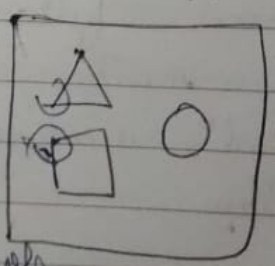
→ add some noise to the image & display the set → loading → filters → noise → display

Write down a python script to demonstrate the use of diff. filters.

↳ 3, 4 are generally used

What are the diff. types of noise available in an image?

↳ Generally used images are Salt & Pepper noises



Edge detection depends on type of noise

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Q1 What do you understand by sampling & quantization of an image?

→ Sampling and Quantization are the two main steps used to convert a continuous (analog) image into a digital image. A digital image is represented as a matrix of pixels, where each pixel stores intensity or color information.

1. Sampling

Sampling is the process of converting the continuous spatial coordinates of an image into discrete pixel locations.

- It determines the spatial ~~distribution~~ resolution of the image (no. of pixels).
- During Sampling, the image is divided into a grid of rows & columns, & each grid cell becomes a pixel.
- A higher sampling rate means more pixels are used, resulting in better image detail and clarity.
- A lower sampling rate ~~means~~ produces fewer pixels, causing loss of detail and a blurred or blocky appearance.

Example:
If an image is sampled into 1024 x 768 pixels, it contains ~~1920~~ 786,432 pixels. Increasing the resolution to 1920 x 1080 pixels captures finer details.

Adv. of Higher Sampling:

- Better image quality
- Fine details are preserved
- Suitable for zooming and printing

Disadv.:-

- Requires more storage space
- Increases processing time

2. Quantization:-

Quantization is the process of converting the continuous intensity (brightness or color) values of sampled pixels into a finite number of discrete levels.

- It determines the intensity resolution of the image.
- Each pixel is assigned the nearest available intensity level.
- The no. of intensity levels depends on the number of bits used per pixel.

Example:

- 8-bit image: $2^8 = 256$ intensity levels (0-255)
- 1-bit image: Only 2 levels (Black & White)
- 16-bit image: 65,536 intensity levels

Higher quantization levels produce smoother images with better contrast, while fewer levels result in loss of detail and visible banding.

Adv. of Higher Quantization:

- Better brightness representation
- Smooth color transitions
- Improved image quality

Disadv.:-

- Large file size
- More memory required

Q? What are the different types of noise available in an image?

→ 1- Gaussian Noise (Amplifier Noise)

- Caused by electronic circuit disturbances during image acquisition.
- Pixel values are changed according to a normal (Gaussian) distribution.

- Appears as fine grain spread throughout the image.

2. Salt-and-Pepper Noise (Impulse Noise)

- Caused by faulty sensors, transmission errors, or memory faults.
- Random pixels become either black (pepper) or white (salt).
- Commonly removed using a median filter.

3. Poisson Noise (Shot Noise)

- Occurs due to the random arrival of photons at the camera sensor.
- Common in low-light & medical imaging.
- Noise level depends on the image brightness.

4. Speckle Noise

- Multiplicative noise mainly found in radar, ultrasound, and SAR images.

- Gives the image a ~~drainy~~ grainy appearance.
- Makes edge detection and image interpretation difficult.

5. Uniform Noise:

- Fixed values are equally likely to be affected within a fixed range.
- Often generated during image digitization or simulation.
- Less common than Gaussian noise.

6. Periodic Noise

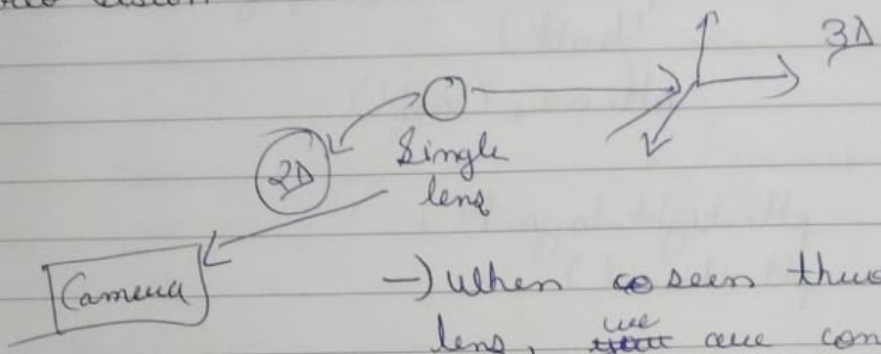
- Caused by electrical or mechanical interference during image capture.
- Appears as repeating patterns or stripes in the image.
- Usually removed using frequency-domain filtering.

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M-2

Vision & Camera
- single lens → Monocular
- Two lens → Stereo

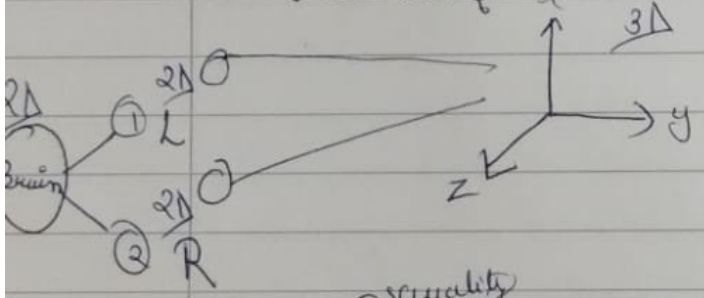
1. Monocular Vision → When you capture something with single eye
2. Stereo Vision



→ When seen through single lens, we are converting an 3Δ image into 2Δ image.

↳ It simulates the human vision.

- Two lens
- There will be displacement b/w the positions of the obj. in the two images captured.



Camera (quality → dist. → focal length)
- lens
  - Intrinsic Features → Focal length
  - Extrinsic Features → Distance,

(Camera Calibration)

Q What do you understand by intrinsic & extrinsic features of camera?

→ Camera should be calibrated with features (both) such as dist., lightning, etc. for image quality.

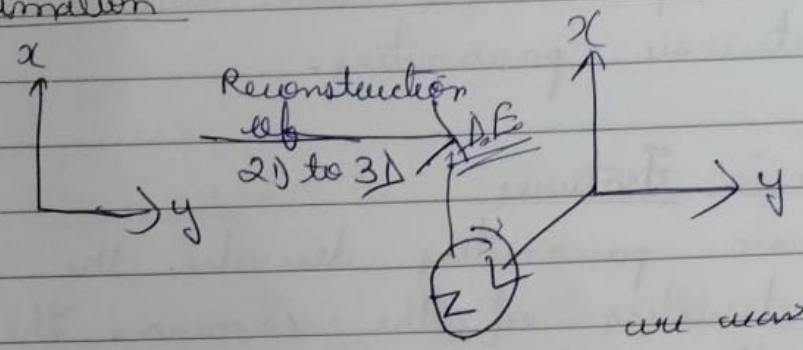
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In terms of Past, Present & Future

Q What are the applications of C.V.?

→ Estimation of the depth of an obj. → Motion

Depth Estimation



→ We get some info., from 2D image, to 3D image, which is known as depth estimation.

We want to correct reconstruction via some measurement

→ Autonomous Vehicle → Stereo Vision

Q. What do you understand by intrinsic & extrinsic features of camera?

→ A camera converts a 3D scene into a 2D image. To understand how this happens, camera parameters are divided into intrinsic & extrinsic parameters.

Intrinsic Features

Intrinsic parameters describe the internal characteristics of the camera. They determine how the camera forms an image & are fixed for a particular camera unless its settings change.

Intrinsic parameters include:

- Focal length (f_x, f_y): Determines the camera's magnification & field of view.
- Principal point (c_x, c_y): The point where the optical axis intersects the image sensor.
- Pixel size / Scale factor: Converts image measurements into pixel coordinates.
- Skew coefficient: Represents the angle b/w the x-axis & y-axis of the image (usually zero in modern cameras).

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- Lens distortion: Includes radial & tangential distortion caused by the camera lens.

Importance:- They are used for camera calibration, image correction, & accurate measurement in computer vision applications.

→ Extrinsic Features:-

Extrinsic parameters describe the position & orientation of the camera with respect to the real-world coordinate system. They define where the camera is located & the direction it is facing.

Extrinsic parameters include:

- Rotation (R): Specifies the orientation of the camera
- Translation (T): Specifies the position of the camera in 3D space.

Importance:- Extrinsic parameters help determine the relationship b/w the camera & objects in the environment. They are widely used in 3D reconstruction, robotics, augmented reality, and navigation.

Q What are the applications of C.V. (in terms of present, past & future)?

→ ~~Computer~~

Past Applications :- ~~In the early~~

- Optical Character Recognition (OCR) for reading printed text.
- Industrial quality inspection to detect defects in manufactured products.
- Barcode & QR code scanning in retail stores.
- Medical image analysis (basic X-ray & CT scan interpretation).
- Satellite image processing for mapping & weather forecasting.

Present Applications :-

- Face recognition for smartphone unlocking & security systems.
- Autonomous vehicles for lane detection, traffic sign recognition, & obstacle detection.

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- Healthcare for disease detection from X-rays, MRI, CT scans, & pathology images.
- Surveillance & security for object tracking & suspicious activity detection.
- Retail for cashier-less stores, inventory management, & customer analytics.
- Agriculture for crop monitoring, weed detection, & disease identification.
- Manufacturing for automated quality control & robotic inspection.
- Augmented Reality (AR) & Virtual Reality (VR) for gaming, education, & entertainment.

Future Applications:-

- Fully autonomous transportation with safer self-driving vehicles.
- Advanced medical diagnosis with faster & more accurate disease prediction.
- Smart cities using intelligent traffic monitoring & public safety systems.
- Human-robot collaboration in industries & homes.

- Precision agriculture using AI-powered drones & robots.
- Personalized education through gesture & emotion recognition.
- Space exploration for autonomous navigation of rovers & satellites.
- Environmental monitoring for wildlife conservation & disaster management.

→ Convolving an Image

For classⁿ of an image

based on CNN → Convolution Neural Network

→ Features

Kernel or Filter

→ Convolution

Stride

Feature extraction

Feature MVP

Classification

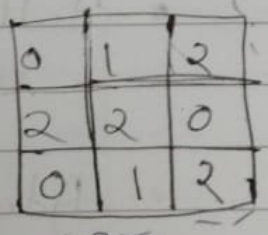
Pooling

Padding

Flatten







$3\times0+3\times1+2\times2+0\times2+0\times2+0\times1+3\times0+1\times1+2\times2 = 12$

$3\times0+3\times1+2\times2+0\times2+0\times2+0\times1+3\times0+1\times1+2\times2 = 12$

$3\times0+3\times1+2\times2+0\times2+0\times2+0\times1+3\times0+1\times1+2\times2 = 12$

$3\times0+3\times1+2\times2+0\times2+0\times2+0\times1+3\times0+1\times1+2\times2 = 12$

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Feature Map

3×3

12	12	17
10	17	19
9	6	14

Stride → Count of steps

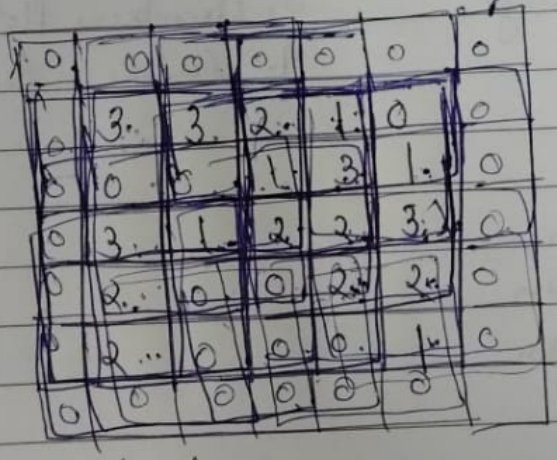
→) Convolving around image with the help of kernel →) Feature extract^{n}

⇒ Same exp^{n} of image in compressed form, reducing computational power →) Feature Map

→ Stride = 2

12 17
9 14

Padding → zeros



3+6+3+2

0	1	2
2	2	0
0	1	2

4+2+3+2 2+1

→ Padding = Stride = 1

Feature Map

6	14	17	11	3
14	12	12	17	11
8	10	17	19	13
14	9	6	14	12
6	4	4	6	4

Mostly used

Max Pooling (2×2)

14	17
11	19

→ Flatten

14
17
11
19

Avg. Pooling → 6+14+14+12 = 46/4 = 11.5

11.5	14.25
9.5	14.0

Computationally complex